Fascial tissue research in sports medicine: from molecules to tissue adaptation, injury and diagnostics

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ABSTRACT

The fascial system builds a three-dimensional continuum of soft, collagen-containing, loose and dense fibrous connective tissue that permeates the body and enables all body systems to operate in an integrated manner. Injuries to the fascial system cause a significant loss of performance in recreational exercise as well as high-performance sports, and could have a potential role in the development and perpetuation of musculoskeletal disorders, including lower back pain. Fascial tissues deserve more detailed attention in the field of sports medicine. A better understanding of their adaptation dynamics to mechanical loading as well as to biochemical conditions promises valuable improvements in terms of injury prevention, athletic performance and sports-related rehabilitation. This consensus statement reflects the state of knowledge regarding the role of fascial tissues in the discipline of sports medicine. It aims to (1) provide an overview of the contemporary state of knowledge regarding the fascial system from the *microlevel* (molecular and cellular responses) to the *macrolevel* (mechanical properties), (2) summarise the responses of the fascial system to altered loading (physical exercise), to injury and other physiological challenges including ageing, (3) outline the methods available to study the fascial system, and (4) highlight the contemporary view of interventions that target fascial tissue in sport and exercise medicine. Advancing this field will require a coordinated effort of researchers and clinicians combining mechanobiology, exercise physiology and improved assessment technologies.

TERMINOLOGY AND DEFINITIONS

The term *fascia* was originally used to describe a sheet or band of soft connective tissue that attaches, surrounds and separates internal organs and skeletal muscles. Advancing research on the physiological and pathophysiological behaviours of a range of connective tissues has revealed that this definition is too restrictive. Understanding of mechanical aspects of connective tissue function depends on consideration of a host of interconnected and interwoven connective tissues beyond these sheets or bands, and there is enormous potential gain from understanding the convergence of biology underpinning adaptation, function and pathology.

> The fascial system includes adipose tissue, adventitia, neurovascular sheaths, aponeuroses, deep and superficial fasciae, dermis, epineurium, joint capsules, ligaments, membranes, meninges, myofascial expansions, periostea, retinacula, septa, tendons (including endotendon/peritendon/epitendon/paratendon), visceral fasciae, and all the intramuscular and intermuscular connective tissues, including endomysium/perimysium/epimysium.[1]

With its diverse components, the fascial system builds a three-dimensional continuum of soft, collagen-containing, loose and dense fibrous connective tissue that permeates the body and enables all body systems to operate in an integrated manner (figure 1).[1] In contrast, the morphological/histological definition describes *fascia* as 'a sheet, or any other dissectible aggregations of connective tissue that forms beneath the skin to attach, enclose, and separate muscles and other internal organs'.[1] The proposed terminology distinguishing the terms 'fascia' and 'fascial system' allows for the precise identification of individual structures as well as grouping them for functional purposes.

CONSENSUS MEETING

The Second International CONNECT Conference was held at the University of Ulm, Germany, on 16–19 March 2017, as part of a conference series aimed at fostering scientific progress towards a better understanding and treatment of fascial tissues in sports medicine. After the conference, a meeting was held with conference speakers and other field-related experts to discuss and find consensus regarding the role of fascial tissue in the field of sports medicine.

Injuries to a variety of fascial tissues cause a significant loss of performance in sports[2] and have a potential role in the development and perpetuation of musculoskeletal disorders, including lower back pain.[3] A major goal of clinicians is to return athletes and patients to activity, training and competition after injury.

This consensus statement reflects the current state of knowledge regarding the role of fascial tissues in the discipline of sports medicine and will be updated as part of a consensus meeting during the CONNECT conference. This paper aims to summarise the contemporary state of knowledge regarding the fascial system from the *microlevel* (molecular and cellular responses) to the *macrolevel* (mechanical properties), and the responses of the fascial system to altered loading (physical exercise), to injury and other physiological challenges including ageing, methods available to study the fascial system, and the contemporary view of

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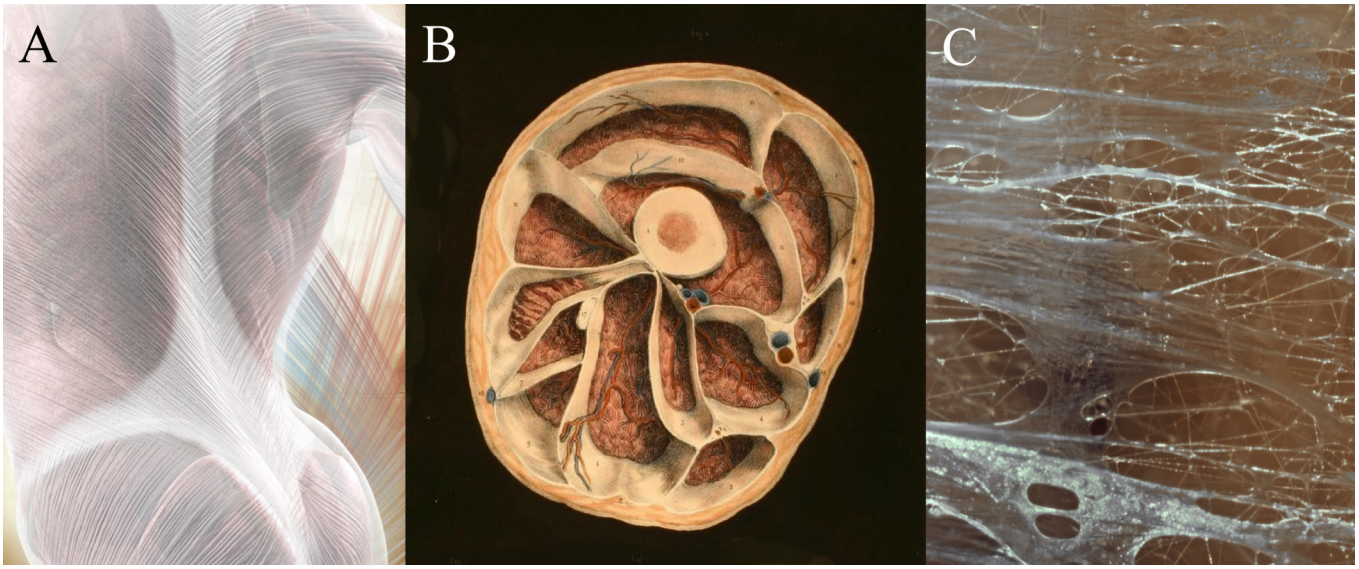

Figure 1 Components of the fascial system. The fascial system includes large aponeuroses like the first layer of the thoracolumbar fascia (A), but also a myriad of enveloping containers around and within skeletal muscles (B) and most other organs of the body. The internal structure of fascial tissues is dominated by collagen fibres which are embedded in a semiliquid ground substance (C). Images with friendly permission from fascialnet.com (A) and thomas-stephan.com (C).

interventions that target fascial tissue in sports medicine. This document was developed for scientists and clinicians to highlight common traps and truths of fascial tissue screening and imaging techniques and intervention methods, and to present a multidisciplinary perspective of future research in the field.

Molecular adaptation of fascial tissues: effects of physical exercise, ageing, sex hormones and inflammation

Molecular crosstalk between extracellular matrix (ECM) molecules and cellular components is an important determinant of fascial tissue physiology and pathophysiology. A molecular chain, characterised by high functional and structural plasticity and bidirectional molecular interactions, connects the cellular cytoskeleton to the ECM (figure 2). Small functional and structural alterations in the ECM result in complex cellular adaptation processes and, vice versa, changes in cell function and structure leading to ECM adaptation.[4] Therefore, fascial tissue homeostasis is the result of a complex interplay and dynamic crosstalk between cellular components and the ECM. Especially under dynamic conditions such as growth and regeneration, strong alterations of the local ECM microenvironments are necessary to allow cellular adaptation and rebuilding of fascial tissues. All factors influencing cell or ECM behaviour can result in changes in the structure and homeostasis of tissues and organs.

The ECM also works as a molecular store, catching and releasing biologically active molecules to regulate tissue and organ function, growth and regeneration. Molecules stored in the ECM network can be cleaved to release biologically active cleavage products.[5] Mechanical stress can induce the release and activation of ECM-stored molecules, inducing the cleavage products of collagen XVIII and other basement membrane components. It has been shown that endostatin (the 20 kDa C-terminal fragment of collagen XVIII) can modulate vascular growth and function.[6–8] In addition, changes in the ECM by ageing or physical exercise may be involved in triggering systemic effects via excreted circulatory molecules, such as the exercise-responsive myokine irisin,[9] which has been proposed to increase energy expenditure in mice and humans.

In fascial tissues such as tendons, acute and chronic loading stimulates collagen remodelling.[10] As the exercise-induced increase in collagen synthesis is lower in women than in men, and

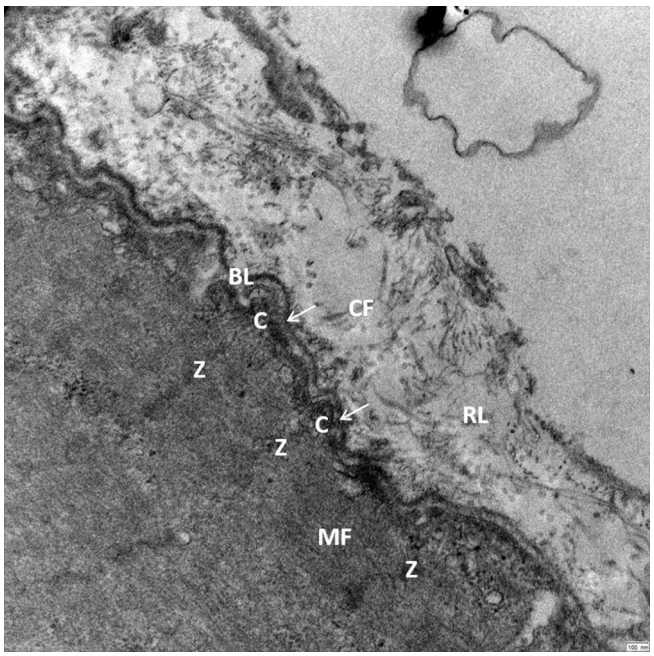

Figure 2 Transmission electron microscopy reveals the close cell–ECM interaction in human skeletal muscle (musculus vastus lateralis, 25 000× magnification) allowing a bidirectional cell–ECM interaction. Myofilaments (MF) are connected by Z-lines (Z) and costameres (C) to the adjacent basal lamina (BL) and the surrounding reticular lamina (RL). Crossbridging structures (arrows) connect the Z-lines and costameres to the dense part of the basal lamina. The reticular lamina is structured by a network of collagen fibrils (CF) and additional ECM molecules, which have a close connection to the basal lamina allowing bidirectional transmission of mechanical forces. ECM, extracellular matrix.

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as injury frequency and the expression of oestrogen receptors in human fascial tissue are sex-dependent, oestrogens may play an important regulatory role in ECM remodelling.[11–13] The effects of oestrogens on collagen synthesis appear to differ between rest and response to exercise. While oestrogen replacement in elderly, postmenopausal women impairs collagen synthesis in response to exercise, oestrogen has a stimulating effect on collagen synthesis at rest.[14] Oral contraceptives, on the other hand, have an overall depressing effect on collagen synthesis.[15]

Physiological ageing is a highly individual process characterised by a progressive degeneration of tissues and organ systems. Age-related alterations in fascial tissues include densification (alterations of loose connective tissue) and fibrosis (alterations of collagen fibrous bundles).[16] Functionally, these pathological changes can modify the mechanical properties of fascial tissues and skeletal muscle, thereby contributing to pain-related and age-related reductions in muscle force or range of motion, which cannot be solely explained by the loss of muscle mass.[17] ECM structural, biochemical, cellular and functional changes occur during ageing.[18] Interestingly, ageing is characterised by chronic, low-grade inflammation—the so-called *inflammaging*.[19] As the ECM is the main site of inflammatory responses taking place in tissues, it is not surprising that the ECM can interact with immune cells to change their function, which is important for growth and regeneration of tissues. Leucocyte extravasation depends on cleavage of the basal membrane by locally released proteases. Tenascin and osteopontin are examples of ECM molecules important for the regulation of the local immune response.[20 21] In addition, ECM plays an important role as a barrier to transmigration of immune cells in and out of the tissue. Although early inflammation after tissue damage due to physical exercise or injury is crucial for tissue remodelling and adaptation,[22 23] stem cell activity and collagen synthesis may be inhibited by the chronic intake of non-steroidal anti-inflammatory drugs prior to exercise.[24 25] However, limiting the magnitude of inflammation might be beneficial for tissue regeneration and gains in muscle mass and strength, depending on the nature of the injury,[26] and in elderly people.[27]

Outlook and perspectives for future research: insights into the structure–function relationship of the ECM, especially in ageing and injured fascial tissues and skeletal muscle, are highly relevant for maintaining musculoskeletal function in the elderly during daily life and exercise and for prevention of exercise-related overuse injuries in athletes. While a body of literature exists on metabolic activity and ECM remodelling in human tendons in response to exercise, much less is known and more research is needed to investigate the molecular response of other fascial tissues (such as intramuscular fascial tissue) to altered loading and ageing.

Myofascial force transmission

Conventionally, skeletal muscles have been considered as primarily transmitting force to their osseous insertions through the myotendinous junction.[28] However, in situ experiments in animals and imaging studies in humans have shown that intermuscular and extramuscular fascial tissues also provide a pathway for force transmission.[29–33] Although the magnitude of non-myotendinous force transmission under in vivo conditions is disputed,[34 35] the contribution of these pathways is thought to be dependent, in part, on the mechanical properties of myofascial tissue linkages.[36] Myofascial tissue that is stiffer or more compliant than normal has been shown to influence the magnitude of intermuscular force transmission and, arguably, may have a significant effect on muscle mechanics.[37–39] The mechanical

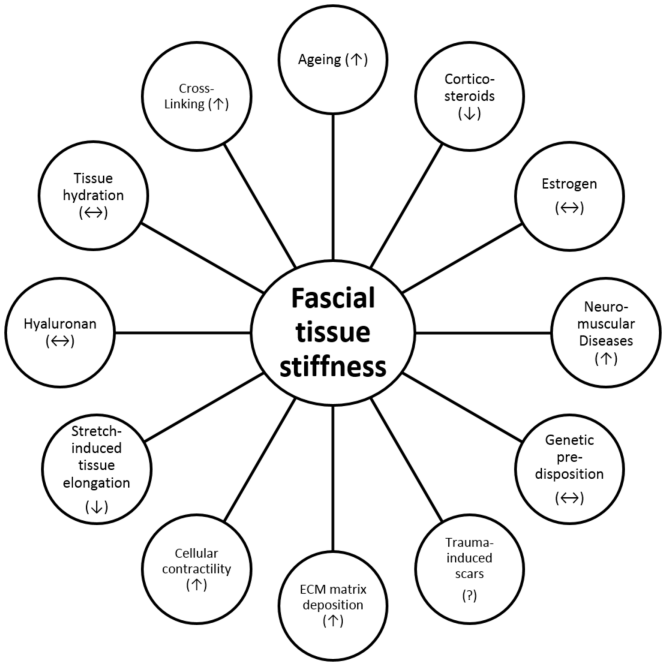

Figure 3 Factors influencing the mechanical stiffness of fascial tissues and their hypothesised impact. Up arrows symbolise a positive effect (eg, increased cellular contractility increases stiffness), down arrows symbolise a negative effect (eg, increased use of corticosteroids decreases stiffness) and double arrows symbolise an ambiguous association (eg, hyaluronan decreases stiffness if mobilised by mechanical stimuli, but leads to increased stiffness if no stimuli are applied). ECM, extracellular matrix.

properties of fascial tissues can be modified by several factors, which, inter alia, include a change in fluid content, crosslinks and molecular organisation and content of specific ECM molecules, and the contractile activity of myofibroblast cells.[40 41] Changes can also be a consequence of muscle injury,[42] disease,[43] surgical treatment[37] or ageing (figure 3).[44]

As fascial tissues connect skeletal muscles, creating a multidirectional network of myofascial continuity,[45] altered local forces (eg, by muscular contraction) might also affect the mechanics of adjacent tissues. In fact, a plethora of cadaveric and animal studies have demonstrated substantial mutual interactions between neighbouring muscles arranged serially in slings (eg, latissimus muscle and gluteus maximus muscle)[46] and parallel to each other (eg, lower limb synergists).[47] For example, when seen from a fascial perspective, the knee-joint capsule is influenced by directly inserting tendons and by more distant structures such as the gluteus maximus or the tensor fasciae latae and their connecting fasciae.[48] However, it remains to be further elucidated how such findings translate into human in vivo conditions.

Although scarce, initial in vivo evidence points towards a significant role of myofascial force transmission for the locomotor system. Available data point towards the existence of (1) remote exercise effects and (2) non-local symptom manifestations in musculoskeletal disorders, both of which might be of relevance in athletic and therapeutic settings. It has been shown that stretching of the lower limb increases the range of motion of the cervical spine, and patients with sacroiliac pain display hyperactivity of the gluteus maximus and the contralateral latissimus muscle.[49–51] Because the involved body regions are connected via myofascial chains, myofascial force transmission might be the cause of the observations. Besides interactions between muscles

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arranged in series, significant amounts of force have been shown to be transmitted in vivo between muscles located parallel to each other; electrical stimulation of the gastrocnemius muscle leads to a simultaneous displacement of the soleus muscle.[30] This intralimb myofascial force transmission may be of relevance in diseases such as cerebral palsy.[38]

Outlook and perspectives for future research: Although the basic mechanisms of myofascial force transmission have been studied, there is a need to discern the influence of variables, such as age, sex, temperature and level of physical activity, within healthy physiological and pathological settings. Furthermore, despite convincing in vitro evidence for the existence of myofascial force transmission, its relative contribution to the occurrence of remote exercise effects under in vivo conditions has to be further elucidated. Besides mechanical interactions between adjacent tissues, non-local changes of stiffness or flexibility may also (at least partly) stem from neural adaptations, for example, a systemic reduction of stretch tolerance.

Injury of fascial tissues: cellular and mechanical responses to damage

Excessive or prolonged loading or direct trauma to fascial tissues initiates micro and macro changes necessary for tissue repair. These effects may also contribute to pathological changes that modify tissue function and mechanics, leading to compromised function of the healthy tissue. Effects may become systemic, and thus not limited to the injured/loaded tissues.

Following an acute injury from overload or anoxia in fascial tissues, the immune response aims to phagocytose injured cells. An acute inflammatory response is typically short-lived and reversible and involves the release of a range of molecules, including proinflammatory cytokines from injured cells and macrophages, along with other substances (eg, bradykinin, substance P and proteases) that sensitise nociceptive afferents[52] and promote immune cell infiltration. If loading is prolonged or repetitive, persistent inflammation may develop,[53 54] leading to the prolonged presence of macrophages and cytotoxic levels of cytokines in and around tissues, ultimately resulting in ongoing tissue damage. Some tissue cytokines (eg, interleukin-1β, tumour necrosis factor (TNF) and transforming growth factor beta (TGFβ-1)) are fibrogenic cytokines that can promote fibrosis via excessive fibroblast proliferation and collagen matrix deposition.[55]

Overproduction of cytokines also maintains sensitisation of nociceptive afferents—a change that would increase production and release of substance P (a known nociceptor neuropeptide). Recent studies show that substance P can stimulate TGFβ-1 production by tendon fibroblasts, and that both substance P and TGFβ-1 can induce fibrogenic processes independently of each other.[56]

Taken together, these findings suggest that both neurogenic processes (nerves are the primary source of substance P) and loading/repair processes (TGFβ-1 is produced by fibroblasts in response to mechanical loading and during repair) can contribute to increased collagen in fascial tissues. Fibrosis (eg, collagen deposition) around the tendon, nerve and myofascial tissues influences dynamic biomechanical properties secondary to tissue adherence and can tether structures to each other or induce chronic compression.[57] Increased collagenous tissues surrounding the nerves can tether the nerves and also enhance pain behaviours.[58] Furthermore, inflammatory cytokines can 'spill over' into the bloodstream, leading to widespread secondary tissue damage and central nociceptor wind-up.[53 59] Circulating TNF is elevated in chronic lower back pain,[60] and recent data highlight a relationship between elevated TNF and greater risk for progression to chronic pain in some individuals[61] and in animal models of overuse.[59]

Muscles also undergo changes in muscle fibre composition, adiposity and fibrosis in response to injury to related structures (eg, injury to an intervertebral disc) even in the absence of muscle trauma (figure 4). These changes closely resemble those identified for direct muscle trauma, such as supraspinatus tendon lesion,[62] although with some differences (eg, differences in the distribution of infiltrating fat). After an injury to an intervertebral disc, deep back muscles undergo rapid atrophy,[63 64] most likely mediated by neural changes such as reflex inhibition.[65] This is followed by changes in muscle fibre composition (slow-to-fast

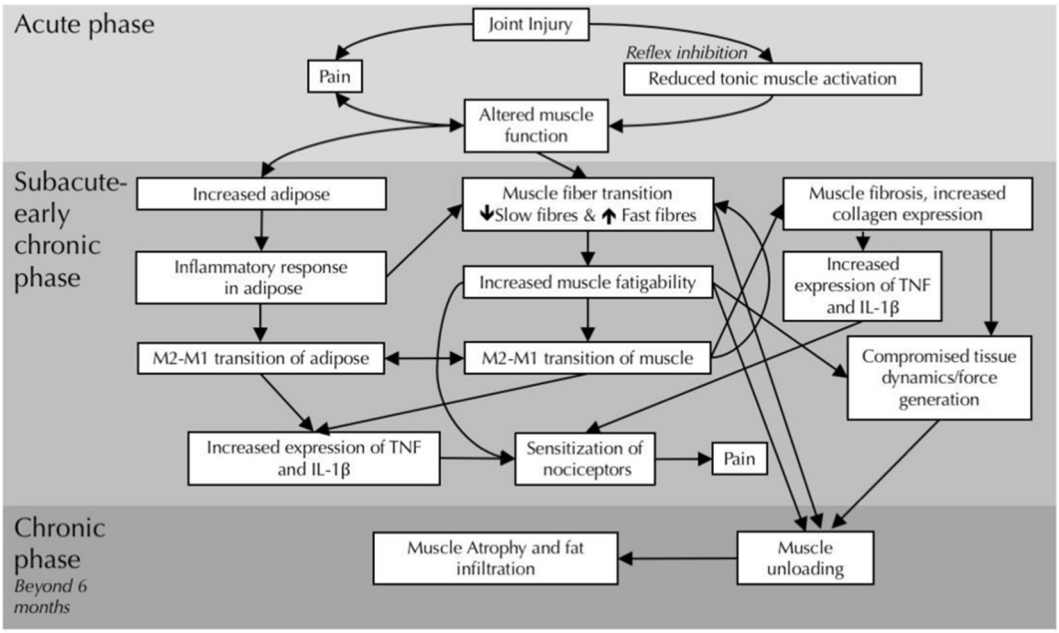

Figure 4 Proposed timeline and mechanisms for fascial, adipose and muscle changes in the multifidus muscle after intervertebral disc lesion. Three phases, acute (top), subacute-early chronic (middle) and chronic (bottom), are characterised by different structural and inflammatory changes. IL-1β, interleukin-1β; TNF, tumour necrosis factor.

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muscle fibre transition), fibrosis and fatty infiltration associated with increased production of proinflammatory cytokines (eg, TNF).[66] Increased cytokine expression was first identified from an mRNA analysis of the muscle, but with an unclear origin. Recent work suggests this is mediated by an increased proportion of proinflammatory macrophages,[67] hypothesised to result from altered metabolic profiles of the muscle as a consequence of transition to more fast (fatigable) muscle fibres.[68] Adipose tissue is a potential source of proinflammatory cytokines and has been implicated in a range of musculoskeletal conditions, including osteoarthritis.[69] Regardless of the underlying mechanism, fibrotic changes in the muscle have a substantial potential impact on tissue dynamics and force generation capacity.

Exercise, physical modalities and pharmacological interventions have all been shown to reduce the inflammatory processes associated with fascial tissue injury and fibrosis. For example, early treatment with anti-inflammatory drugs can prevent/reverse pain behaviours induced by TNF signalling and reduce downstream collagen production in animal models.[70] Stretching of fascial tissues can promote resolution of inflammation both in vivo and in vitro,[71] and manual therapy can prevent overuse-induced fibrosis in several fascial tissues.[72] In terms of muscle changes, resistance exercise is necessary to reverse fatty changes (and perhaps fibrosis) in chronic conditions,[73] whereas gentle muscle activation is sufficient to reverse early muscle atrophy,[74] and whole body exercise can prevent inflammatory changes in back muscles that follow intervertebral disc injuries.[75]

Outlook and perspectives for future research: Future research is needed to gain a deeper understanding of the mechanisms underlying the impact of treatments on fibrosis and fatty changes in fascial tissues. Although there is evidence that exercise, physical therapies or pharmacological approaches can impact inflammatory processes, and reduce consequences, further work is required to understand how best to tailor interventions based on the time-course of pathology and type of exercise, or whether there is additional benefit from combined treatments.

Imaging and non-imaging tools for diagnosis and assessment

Pathological changes in the mechanical properties of fascial tissues have been hypothesised to play an essential role in musculoskeletal disorders such as chronic pain conditions and overuse injuries.[76] As a result, considerable demand for diagnostic methods examining fascial tissue function has arisen. In basic research, an oft-used approach is to study molecular and mechanical changes in myofibroblasts and other biomarkers via needle biopsy and subsequent immunohistochemistry.[77]

To evaluate the effects of treatment and exercise in clinical settings, a series of methods are available (table 1). Changes in water content can be analysed via bioimpedance assessment,[78] but there are no data on reliability and validity of measurements in smaller body regions. Manual palpation represents a cost-neutral and widely used screening method aimed at assessing viscoelastic properties (eg, stiffness); however, similarly, its reliability is limited.[48 79 80 81] However, the approach is based on a number

Table 1 Currently used diagnostic methods to examine fascial tissue structure and function

Method	Assessment target	Advantages	Disadvantages	References
Biopsy	Histological properties including molecular analysis.	Permits analysis of tissue damage, infiltration of inflammatory cells, cytokines and others.	Invasiveness.	66 75 77
Bioimpedance	Hydration changes.	High sensitivity.	Lacking data on reliability and validity for smaller regions.	78
Manual palpation	Stiffness, elasticity and shearing mobility of tissue.	Cost-effectiveness. Psychosocial factors.	Limited reliability.	79 80 82
Indentometry	Stiffness and elasticity.	Established reproducibility.	Limited depth.	) 81 83–85
Ultrasound (US) imaging	Thickness of layers, tendon elongation.	Permits diagnosis of a fibrotic thickening (eg, of a particular endomysium) or of tendon strain response during loading.	Difficulty in standardising the exact viewing angle.	86 88
US with correlation software	Relative shearing motion of adjacent layers.	Permits diagnosis of adhesive tissue connections, such as in chronic low back pain.	Lacking standards for selection of regions of interest.	89
Compression-based US elastography	Stiffness.	Measurements possible at further depth than, for example, with indentometry.	Lack of standardisation. Frequent appearance of artefacts.	87
Shear-wave US elastography	Stiffness.	Enhancement by propagation analysis permits morphological analysis.	Lack of standardisation.	90 91
B-mode ultrasonography	Tendon structure and mechanical/material properties.	1. In vivo methodology. 2. Application in perspective studies. 3. Relatively inexpensive.	1. Accuracy is user-dependent. 2. Applicability is limited to superficial tendons mainly. 3. Limited control of any mediolateral deviation of the tendon line of pull off the scanning plane. 4. Tendon slack length (ie, at 0% strain) and tendon force cannot be directly measured and need to be estimated. 5. Scanning frame rate is currently limited.	90 96–98 103

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of assumptions, and available devices often lack a thorough proof of validity.[77 82] Moreover, no tissue-specific conclusions can be drawn due to the black-box character of the measurements.[83] Imaging methods such as ultrasound or elastography, in contrast, are promising tools for explicitly quantifying the mechanical properties of fascial tissues under in vivo conditions.[84]

Producing a distortion of the measured tissue (eg, through compression or shear waves), elastography provides ultrasound images reflecting the relative hardness of the targeted area. Recently, the technique has been increasingly applied in musculoskeletal research. However, the existence of several different methods, lack of standardisation and frequent appearance of artefacts during measurements threaten the validity of achieved results.[85] Without the use of elastography, the conventional ultrasound image can be reliably used to display and measure the morphology of fascial tissues, such as myofascial tissues, ligaments and tendons.[86] Some initial studies have, moreover, attempted to quantify relative movement (eg, sliding of fascial layers and shear strain) using cross-correlation calculations.[87]

Despite some initial applications to myofascial tissues, most data on ultrasound imaging are available for tendon measurements (figure 5). In the late 1990s, advancements made in the application of B-mode ultrasonography allowed quantification of the tensile deformation of human tendons, in vivo, based on tracking of anatomical features in the tendon when pulled on by the force exerted in the in-series muscle during static contraction.[88] Unfortunately, the in vivo stiffness and Young's modulus results often disagree with findings from in vitro material tests, when forces and elongations are precisely controlled and measured. Errors are likely being caused by in vivo measurement simplifications in the quantification of both tendon deformation and the loading applied during the static muscle contraction. The former includes simplifications regarding the tendon's resting length, line of pull and uniformity in material properties. The latter includes simplifications regarding the effect of loading on tendon moment arm length, the effect of antagonist muscle coactivation and the uniformity in tendon cross-sectional area. Most of these simplifications can be avoided by appropriate measurements to quantify the neglected effects. In addition, recent developments in ultrasound shear-wave propagation[89] and speckle tracking[90] have the potential to substantially improve experimental accuracy and physiological relevance of in vivo findings.

In contrast to static muscle contraction tests aimed at assessing human tendon stiffness and Young's modulus, scanning during dynamic activities has typically been applied to document tendon deformations directly, through morphometric analysis on scans,[90 91] or indirectly, through ultrasound propagation speed analysis,[92 93] to investigate the interaction between tendon and muscle in the studied task. These experimental approaches are relatively immune to problems caused by erroneous quantification of tendon forces; however, appropriate measurements need to be taken to validate the assumption that the usual practice of tracking a single tendon anatomical point, or a tendon region limited by the size of the scanning probe, can give a representative picture for the entire tendon.

Outlook and perspectives for future research: In view of the current diagnostic methods' limitations, further research investigating the measurement properties (eg, validity) is warranted to provide evidence-based recommendations. Hence, within the clinical assessment of mechanical soft-tissue properties, collected data should be interpreted with caution, and, as long as no clear gold standards exist, a combination of methods seems advisable instead of focusing exclusively on one technique. Ultrasound-based assessments of tendon deformability on loading have grown in popularity but can provide erroneous conclusions due to several invalid assumptions and approximations typically made to simplify the experimental protocol. Most of these errors can be eliminated by appropriate measurements.

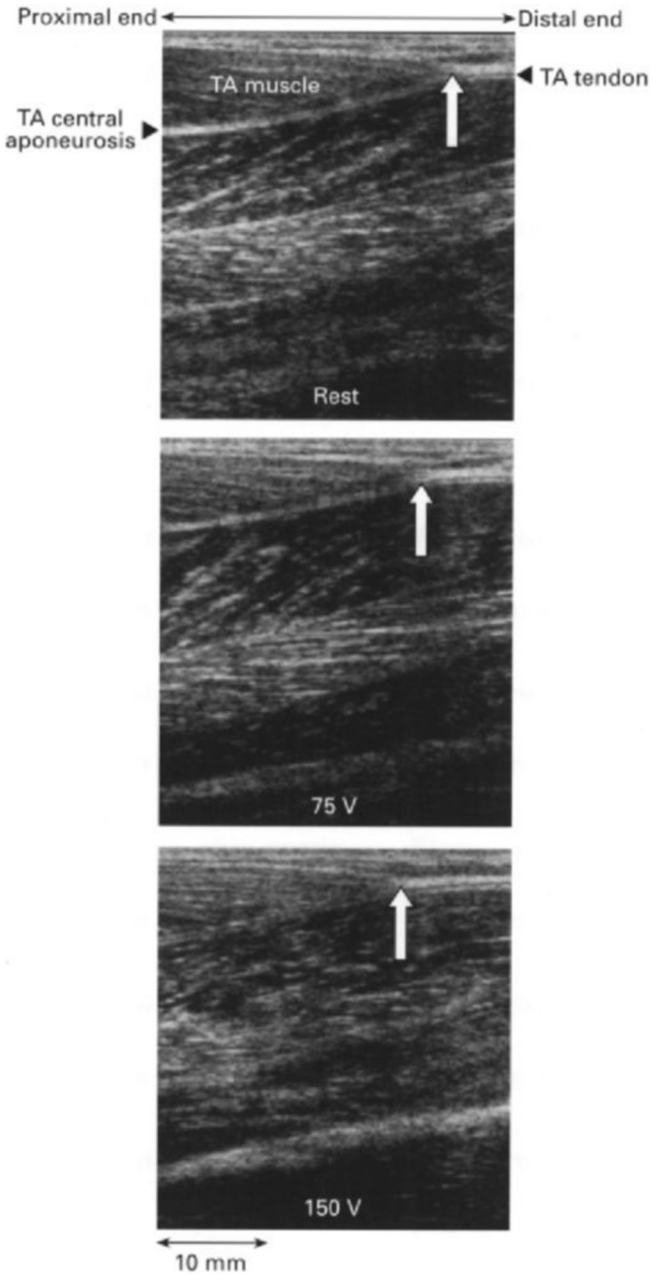

Figure 5 Tendon displacement measured by B-mode ultrasound. Sonographic images of the human tibialis anterior (TA) muscle at rest (top) and in response to electrical stimulation at 75 V (middle) and 150 V (bottom). The white arrow indicates the TA tendon origin. Notice the proximal shift of the TA tendon origin on electrical stimulation.[88]

Mechanobiology of fascial tissues: effects of exercise and disuse

The main principles of the above ultrasound-based methodology have been implemented in numerous studies over the last 20 years to study the adaptability of human tendons to exercise and disuse.[94 95] The findings convincingly show that human tendons respond to the application of chronic overloading by increasing their stiffness and to chronic unloading by decreasing

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their stiffness. The mechanisms underpinning these adaptations include changes in tendon size and changes in Young's modulus. One common finding among studies is that tendon adaptations occur quickly, within weeks of mechanical loading/unloading application.[96 97] Importantly, however, some studies report adaptations in tendon size but not tendon material,[98] and others in tendon material but not size,[96] while some report adaptations in both tendon size and material.[99]

To study human tendon mechanobiology and explore the basis of the above distinct adaptability features, both cross-sectional and longitudinal experimental designs have often been adopted. Cross-sectional designs have been used for the following purposes: (1) to compare tendons subjected to different habitual loads due to their specific anatomical location,[100] (2) to compare tendons between limbs with muscle strength asymmetry,[98] (3) to compare tendons in humans with different body mass but similar habitual activities[95] and (4) to compare tendons in athletes with those in sedentary individuals.[99] Study designs (1), (2) and (3) support the notion that adjustments in tendon stiffness to accommodate changes in physiological loading are accomplished by adding or removing tendon material rather than altering Young's modulus of the tendon. Importantly, the addition or removal of tendon material does not seem to always occur uniformly along the tendon, but in some regions only, which can go undetected unless the whole tendon is examined.[101] In contrast to study designs (1), (2) and (3), findings from study design (4) show that improvements in Young's modulus of the tendon may occur and account fully for, or contribute to, the increased tendon stiffness in response to loading. Interestingly, exercise-training intervention studies also report improvements in Young's modulus of the tendon.[94–96] In combination, these findings indicate that stiffening of the tendon through alteration of its material requires 'supra-physiological' loading features (eg, in terms of loading magnitude, frequency and/or duration). Once this rapid adaptation occurs and the exercise becomes a habitual daily activity, alterations in tendon size might mediate any further changes in tendon stiffness.

Outlook and perspectives for future research: Combining ultrasonography with dynamometry methods has now made it possible to assess in vivo human tendon plasticity under conditions of altered mechanical loading. Two important questions warrant further research. (1) What is the mechanism underpinning regional differences in tendon adaptability in terms of tendon size? Possibilities worth investigating include differences in local stress, local Young's modulus, local blood flow and mechanotransduction sensitivity. Finite element modelling of the tendon may be an appropriate avenue to examine the first two possibilities. (2) What is the limiting factor in tendon plasticity to exercise? An intuitive answer is that the magnitude and time-course of tendon plasticity are merely determined by how much and how fast the in-series muscle force increases as the muscle adapts to the chronically increased load, but confirming this requires systematic research.

Interventions for fascial tissue pathologies in sports medicine

Fascial tissue dysfunction in the field of sports medicine is rarely treated surgically. Anti-inflammatory drugs are used for sports-related overuse pathologies; however, they may impair regeneration and diminish tissue adaptation.[24 25] Gyrase-inhibiting antibiotics often contribute to an increased likelihood of tendon injuries in sports.[102] In addition, injections of platelet-rich plasma seem to be successful in some cases of tendinopathy, although efficacy remains inconclusive.[67 103] Moderate evidence exists on the value of shockwave therapy and eccentric loading in tendon healing.[104 105] Similarly, foam rolling (tool-assisted massage of myofascial tissues) seems to improve short-term flexibility and recovery from muscle soreness[75 106 107] and decrease latent trigger point sensitivity.[103] Nevertheless, the physiological mechanisms of these reported effects remain unclear, although initial evidence suggests increases in arterial perfusion, enhanced fascial layer sliding and modified corticospinal excitability following treatment[108 109] (F Krause *et al*, submitted, 2018). Finally, manual therapies, such as massage, osteopathy or Rolfing (a massage technique based on achieving symmetrical alignment of the body), are frequently used to improve fascial tissue regeneration or athletic performance, although their efficacy still remains to be validated.[110 111]

Outlook and perspectives for future research: Hopefully, current and future improvements in assessment methodologies will generate more conclusive research regarding which treatment modalities are most promising for specific conditions. While commercial and other interests often favour the promotion of premature positive conclusions about specific fascia-related treatments, strict application of scientific rigour is essential for the development of this promising field.

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